

Grid L

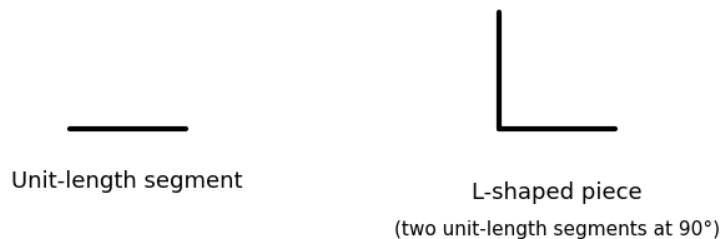
time limit per test

2 seconds

memory limit per test

256 megabytes

Roger has  $p$  unit-length segments and  $q$  L-shaped pieces, each formed by joining two unit-length segments at a right angle.



He wants to use all of these pieces, with no piece left unused, to form a grid of dimensions  $n \times m$ .

Given  $p$  and  $q$ , determine whether there exist positive integers  $n$  and  $m$  such that an  $n \times m$  grid can be formed using exactly  $p$  unit-length segments and  $q$  L-shaped pieces, possibly rotating them, and provide any valid  $n$  and  $m$ .

### Input

Each test contains multiple test cases. The first line contains the number of test cases  $t$  ( $1 \leq t \leq 100$ ). The description of the test cases follows.

The first line of each test case contains two integers  $p$  and  $q$  ( $1 \leq p, q \leq 108$ ).

### Output

For each test case, print any valid  $n$  and  $m$  such that it's possible to construct  $n \times m$  using all pieces. If there are no such  $n$  and  $m$ , print  $-1$ .

### Example

#### Input

7

1 2

1 3

5 1

2 5

2 10

100000000 100000000

1 1

### Output

-1

1 2

1 2

2 2

2 4

-1

-1

### Note

Here are the constructions for examples 2, 3, and 4. We use a different color for different L-shaped pieces, black for all unit-length segments.

